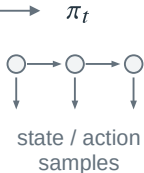


1 Rollout



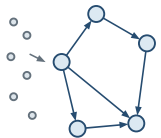
GAE

$$A_r, A_{c_k}$$

observed episode cost

$$\bar{J}_k$$

2 Flow inputs



edge attributes

$$\hat{F} \quad A \quad C_k$$

advantages

3 Constraints

soft budgets

$$\langle C_k, F \rangle \leq b_k$$



$$b_k = B_k / L$$

hard exclusion

$$\bigcirc \text{--}\text{X}\text{--}\bigcirc F_H = 0$$

4 Entropic projection

$$\min_{F \geq 0} [-\langle A, F \rangle + \varepsilon \text{KL}(F \| \hat{F})]$$

s.t. flow balance + budgets



Target flow is not executed

budget duals

$$\bar{\lambda}_k^* \leftarrow \text{clip}_c + \text{EMA} \leftarrow \lambda_k^*$$

5 Priced policy update

$$\beta_k \leftarrow [\beta_k + \alpha(\bar{J}_k - B_k)]_+$$



$$\lambda_k$$

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO/TRPO $\rightarrow \pi_{t+1}$

next rollout